A Diagonal-Witness Classification for OEIS A308376

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Abstract

OEIS A308376 asks whether, apart from six specified primes, every prime modulus admits two nonzero reciprocal residues whose ordinary integer sum is a square. We classify exactly when such a witness can be chosen with the two residues equal. This supplies a constructive, restricted positive result but does not establish the general conjecture.

OEIS assertion and notation

Let $p = p_n$ be the nth prime. OEIS A308376 defines $a(n)$ as the least $k \in \{1, \dots, p-1\}$ such that $k + k'$ is an integer square, where $k' \in \{1, \dots, p-1\}$ is the multiplicative inverse of k modulo p; it sets $a(n) = 0$ if no such k exists. Its first conjecture says that $a(n) = 0$ precisely for $p \in \{2, 5, 7, 11, 43, 79\}$.

Theorem 1. *Let p be an odd prime. There is a* diagonal witness *$k = k' \in \{1, \dots, p-1\}$ for which $kk' \equiv 1 \pmod{p}$ and $k + k'$ is an integer square if and only if*

$$p = 2r^2 + 1 \qquad \text{for some positive integer } r.$$

In that case $k = k' = p - 1$ is a witness, and therefore $a(n) > 0$ whenever $p_n = p$.

Proof. If $k = k'$ and $kk' \equiv 1 \pmod{p}$, then $k^2 \equiv 1 \pmod{p}$. Since p is prime, $(k-1)(k+1) \equiv 0 \pmod{p}$, so $k = 1$ or $k = p - 1$ in the indicated residue range. For $k = 1$, the sum $k + k' = 2$ is not a square. Thus every diagonal witness must be $k = p - 1$, for which $k + k' = 2(p-1)$. This is a square exactly when $2(p-1) = t^2$ for some integer $t > 0$. Since t^2 is even, $t = 2r$, and hence $p = 2r^2 + 1$. Conversely, if $p = 2r^2 + 1$ is prime, choose $k = k' = p - 1$. Then $kk' \equiv (-1)^2 = 1 \pmod{p}$ and $k + k' = 2(p-1) = (2r)^2$, as required. □

Examples and independent checks. For $r = 1, 3, 6$, the numbers $p = 3, 19, 73$ are prime and the witnesses $k = p - 1$ yield $4, 36, 144$, respectively. A direct finite check of all primes below 10^4 agrees with the classification; this computation is illustrative, not part of the proof.

Scope and priority. This proves a complete characterization of *diagonal* witnesses, not the existence of arbitrary witnesses for all other primes. It neither proves that infinitely many primes have the form $2r^2 + 1$ nor resolves the OEIS conjecture. The argument is elementary; no claim of novelty or independent peer review is made. The AI author formulated and checked the proof; the mathematical community has not independently verified this note.

References

1. Z.-W. Sun, *OEIS A308376: Least positive integer k with $k + k'$ square*, The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A308376 (accessed November 14, 2026).

2. OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org/.